

Project Questions & Answers (SQL Queries)

-- 1. Sales & Growth Performance

-- first question: What is the Total Sales amount for 1997 and 1998 combined?

-- Answer:

```
SELECT SUM(PD.product_retail_price * SF.quantity ) AS Total_Sales
FROM Sales_FACT AS SF
LEFT JOIN Products_DIM AS PD
ON SF.product_id = PD.product_id
```

-- 1. Sales & Growth Performance

-- Second question: What are the Total Orders count?

---Answer:

```
SELECT SUM ( Total_orders_per_customer) AS Total_no_of_orders
FROM (
    SELECT customer_id ,
           COUNT ( DISTINCT transaction_date ) AS Total_orders_per_customer
    FROM Sales_FACT
    GROUP BY customer_id
) AS Customer_Order
```

-- 1. Sales & Growth Performance

-- Third question : what is the Average Order Value (AOV)?

--- Answer:

```
SELECT AVG( Sales_per_order ) AS Average_order_value
FROM (
    SELECT SF.customer_id ,
           SF.transaction_date,
           SUM(PD.product_retail_price * SF.quantity ) AS Sales_per_order
    FROM Sales_FACT AS SF
    LEFT JOIN Products_DIM AS PD
    ON SF.product_id = PD.product_id
    GROUP BY SF.customer_id,SF.transaction_date
) AS Order_Detail
```

-- 2. Product & Quality Insights

-- First question : Which Product Brands are the "Top 5" by revenue?

--- Answer:

```
SELECT TOP 5 PD.product_brand,
            SUM(PD.product_retail_price*SF.quantity) AS Sales_by_brand
FROM Sales_FACT AS SF
JOIN Products_DIM AS PD
ON SF.product_id = PD.product_id
GROUP BY PD.product_brand
ORDER BY Sales_by_brand DESC
```

-- 2. Product & Quality Insights

-- Second question : What are the Total Returns count?

--- Answer:

```
SELECT SUM(quantity) AS Total_returns
FROM Returns_FACT
```

-- 2. Product & Quality Insights

-- Third question : What is the overall Return Rate % per brand?

--- Answer: USING CTE

```
WITH BD AS (
    SELECT SB.product_brand , SB.SALES_PER_BRAND ,
    RB.RETURNS_PER_BRAND
    FROM ( SELECT PD.product_brand,
                SUM(SF.quantity) AS SALES_PER_BRAND
            FROM Sales_FACT AS SF
            LEFT JOIN Products_DIM AS PD
            ON SF.product_id = PD.product_id
            GROUP BY PD.product_brand ) AS SB
    LEFT JOIN
        ( SELECT PD.product_brand,
                SUM(RF.quantity) AS RETURNS_PER_BRAND
            FROM Returns_FACT AS RF
            LEFT JOIN Products_DIM AS PD
            ON RF.product_id = PD.product_id
            GROUP BY PD.product_brand ) AS RB
    ON SB.product_brand=RB.product_brand
)
SELECT BD.product_brand,
       BD.SALES_PER_BRAND,
       BD.RETURNS_PER_BRAND,
       ((BD.RETURNS_PER_BRAND*100.0)/BD.SALES_PER_BRAND) AS RETURN_RATE
FROM BD
ORDER BY RETURN_RATE DESC
```

-- 3. Customer segmentation

--First Question: How does Yearly Income correlate with the total quantity of products purchased?

--- Answer:

```
SELECT CD.yearly_income,
       SUM(SF.quantity) AS T_Prod_quantity,
       COUNT (DISTINCT CONCAT (SF.transaction_date,SF.customer_id )) AS TOTAL_TRANS,
       CAST ( SUM(SF.quantity)*1.0 / COUNT (DISTINCT CONCAT
(SF.transaction_date,SF.customer_id )) AS DECIMAL (10,2) )AS ITEMS_PER_ORDER
FROM     Sales_FACT      AS SF
LEFT JOIN Customers_DIM AS CD
ON       SF.customer_id = CD.customer_id
GROUP BY CD.yearly_income
order by T_Prod_quantity DESC
```

-- 3. Customer segmentation

-- Second question : --Do customers with specific Member Cards have a higher lifetime value?

-- Answer:

-- Total revenue & AVG LIFETIME VALUE by member card

```
SELECT CD_DET.member_card,
       COUNT (CD_DET.customer_id) AS NO_OF_CUST,
       SUM(CD_DET.TOTAL_SALES)     AS T_SALES,
       SUM(CD_DET.TOTAL_SALES) / COUNT (CD_DET.customer_id) AS AVG_LTV
FROM (
    SELECT SF.customer_id,
           CONCAT ( CD.first_name , ' ', CD.last_name) AS FULL_NAME ,
           CD.member_card,
           CD.education,
           COUNT ( Distinct SF.transaction_date )      AS TOT_TRANSACTIONS,
           SUM( PD.product_retail_price * SF.quantity ) AS TOTAL_SALES
    FROM     Sales_FACT      AS SF
    LEFT JOIN Products_DIM   AS PD ON SF.product_id = PD.product_id
    LEFT JOIN Customers_DIM  AS CD ON SF.customer_id = CD.customer_id
    GROUP BY SF.customer_id,
           CONCAT ( CD.first_name , ' ', CD.last_name),
           CD.member_card,
           CD.education
) AS CD_DET
GROUP BY CD_DET.member_card
ORDER BY AVG_LTV DESC
```

-- 4. Regional & Store Efficiency

-- First question: Are "Supermarkets" more profitable than "Small Grocery" stores?

-- Answer:

```
SELECT      SD.store_type,
            COUNT (DISTINCT SF.store_id)           AS TOT_STORES,
            SUM( SF.quantity*PD.product_retail_price) AS TOT_REVENUE,
            SUM( SF.quantity*PD.product_cost)        AS TOT_COST,
            SUM( SF.quantity*PD.product_retail_price) - SUM( SF.quantity*PD.product_cost)
AS TOT_PROFIT,
            CAST ((SUM( SF.quantity*PD.product_retail_price) - SUM(
SF.quantity*PD.product_cost))*100 / SUM( SF.quantity*PD.product_retail_price) AS DECIMAL
(10,2) ) AS PROFIT_MARGIN,
            CAST ((SUM( SF.quantity*PD.product_retail_price) - SUM(
SF.quantity*PD.product_cost))/ COUNT (DISTINCT SF.store_id) AS decimal (10,2))AS
AVG_PROFIT_PER_STORE
FROM        Sales_FACT AS SF
LEFT JOIN   Stores_DIM AS SD ON SF.store_id = SD.store_id
LEFT JOIN   Products_DIM AS PD ON SF.product_id = PD.product_id
GROUP BY    SD.store_type
```

-- 4. Regional & Store Efficiency

-- Second question: What are the Sales per Square Foot across different store sizes?

-- Answer:

```
SELECT      SD.store_id,
            SD.store_type,
            SD.total_sqft,
            SD.grocery_sqft,
            CAST( (SD.grocery_sqft*100.0) / SD.total_sqft AS decimal (10,2) ) AS
Utilization_Ratio,
            SUM(SF.quantity*PD.product_retail_price) AS STORE_TOT_SALES ,
            CAST (SUM(SF.quantity*PD.product_retail_price) / SD.total_sqft AS DECIMAL
(10,2) )AS SALES_PER_TOT_Sqft,
            CAST (SUM(SF.quantity*PD.product_retail_price) / SD.grocery_sqft AS DECIMAL
(10,2) )AS SALES_PER_GROCERY_Sqft
FROM        Sales_FACT AS SF
LEFT JOIN   Stores_DIM AS SD ON SF.store_id = SD.store_id
LEFT JOIN   Products_DIM AS PD ON SF.product_id = PD.product_id
GROUP BY    SD.store_id,SD.store_type,SD.total_sqft,SD.grocery_sqft
ORDER BY    Utilization_Ratio ASC
```

-- 5. Supply Chain Timing

-- First question : What is the average time between the stock date and the transaction date?

-- Answer:

```
SELECT
            CAST(AVG(DATEDIFF(day, stock_date, transaction_date)*1.0) AS DECIMAL (10,2) )AS
AVG_DAYS_ON_SHELF
FROM Sales_FACT;
```